

# Business KPI Investigation Report

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**Dataset:** business\_data  
**Table:** sales

## KPI Summary

| Metric                          | Value               |
|---------------------------------|---------------------|
| total_revenue                   | 241415.960000000017 |
| total_orders                    | 1000                |
| total_quantity_sold             | 4518                |
| average_order_value             | 241.415960000000015 |
| average_discount_rate           | 0.06274999999999999 |
| average_shipping_cost_per_order | 4.56586             |

## Business Analytics

- The business generated a total revenue of \$241,415.96 from 1,000 orders.
- A total of 4,518 units were sold across all orders, indicating an average of approximately 4.5 units per order.
- The average order value (AOV) stands at \$241.42, suggesting a healthy transaction size per customer.
- The average discount rate applied across orders was 6.27%, which could be a strategy to drive sales volume or attract new customers.
- The average shipping cost per order was \$4.57, representing an operational expense that should be monitored for efficiency.

## Anomalies

- The average shipping cost per order (\$4.57) appears unusually low given the average order value (\$241.42) and average units per order (4.5), potentially indicating highly efficient logistics, significant shipping subsidies, or a product mix of exceptionally small/light items that warrants further investigation into its sustainability and true cost.

## Root Causes

- Highly optimized logistics and supply chain operations (e.g., efficient routing, bulk shipping, localized fulfillment centers).
- Strategic business decision to heavily subsidize or absorb shipping costs (e.g., promotional free shipping, customer acquisition strategy).
- Product portfolio primarily consists of small, lightweight, or low-cost-to-ship items.
- Favorable long-term contracts and bulk shipping rates negotiated with carriers.

- Potential miscategorization or incomplete tracking of actual shipping expenses within the financial reporting system, leading to an understated KPI.

## Recommendations

- Prioritize an immediate audit of shipping cost data and accounting practices to confirm that all relevant expenses (e.g., carrier fees, packaging, labor for fulfillment, insurance) are accurately captured and attributed to shipping, addressing the potential for miscategorization.
- If shipping costs are intentionally subsidized (e.g., free shipping promotions), conduct a comprehensive ROI analysis to evaluate their impact on customer acquisition, retention, average order value, and overall profitability, ensuring strategic alignment with business goals.
- Identify and document the specific operational efficiencies or favorable carrier contract terms that contribute to the unusually low shipping costs. Explore opportunities to further enhance these efficiencies and consider leveraging them as a competitive differentiator in marketing and sales.
- Based on the data verification and ROI analysis, evaluate alternative shipping models such as tiered pricing, minimum order values for free shipping, or adjusting product bundling strategies to optimize profitability without compromising customer satisfaction.
- Analyze the current product mix's influence on shipping costs and assess how potential changes in product offerings (e.g., introducing larger or heavier items) might impact future logistics expenses, allowing for proactive planning and strategy adjustments.

## Detailed AI Investigation Report

### Business Investigation Report

#### \*\*1. Executive Summary\*\*

This report presents an analysis of recent business performance, highlighting key operational metrics and identifying a significant anomaly related to shipping costs. The business generated a total revenue of \$241,415.96 from 1,000 orders, with a healthy average order value of \$241.42. A total of 4,518 units were sold, indicating an average of 4.5 units per order. The average discount rate applied was 6.27%. The primary anomaly identified is an unusually low average shipping cost per order of \$4.57, which warrants immediate investigation. Potential root causes range from highly optimized logistics and strategic subsidies to potential data miscategorization. Recommendations focus on a comprehensive audit of shipping costs, ROI analysis of subsidies, documentation of efficiencies, evaluation of alternative shipping models, and analysis of product mix impact. Addressing this anomaly is crucial for ensuring accurate financial reporting and optimizing operational profitability.

#### \*\*2. KPI Summary\*\*

\* \*\*Total Revenue:\*\* \$241,415.96  
\* \*\*Total Orders:\*\* 1,000  
\* \*\*Total Quantity Sold:\*\* 4,518  
\* \*\*Average Order Value (AOV):\*\* \$241.42  
\* \*\*Average Discount Rate:\*\* 6.27%  
\* \*\*Average Shipping Cost Per Order:\*\* \$4.57

#### \*\*3. Business Analytics\*\*

- \* The business generated a total revenue of \$241,415.96 from 1,000 orders.
- \* A total of 4,518 units were sold across all orders, indicating an average of approximately 4.5 units per order.
- \* The average order value (AOV) stands at \$241.42, suggesting a healthy transaction size per

customer.

- \* The average discount rate applied across orders was 6.27%, which could be a strategy to drive sales volume or attract new customers.

- \* The average shipping cost per order was \$4.57, representing an operational expense that should be monitored for efficiency.

#### **\*\*4. Anomalies\*\***

- \* The average shipping cost per order (\$4.57) appears unusually low given the average order value (\$241.42) and average units per order (4.5), potentially indicating highly efficient logistics, significant shipping subsidies, or a product mix of exceptionally small/light items that warrants further investigation into its sustainability and true cost.

#### **\*\*5. Root Causes\*\***

- \* Highly optimized logistics and supply chain operations (e.g., efficient routing, bulk shipping, localized fulfillment centers).

- \* Strategic business decision to heavily subsidize or absorb shipping costs (e.g., promotional free shipping, customer acquisition strategy).

- \* Product portfolio primarily consists of small, lightweight, or low-cost-to-ship items.

- \* Favorable long-term contracts and bulk shipping rates negotiated with carriers.

- \* Potential miscategorization or incomplete tracking of actual shipping expenses within the financial reporting system, leading to an understated KPI.

#### **\*\*6. Recommendations\*\***

- \* Prioritize an immediate audit of shipping cost data and accounting practices to confirm that all relevant expenses (e.g., carrier fees, packaging, labor for fulfillment, insurance) are accurately captured and attributed to shipping, addressing the potential for miscategorization.

- \* If shipping costs are intentionally subsidized (e.g., free shipping promotions), conduct a comprehensive ROI analysis to evaluate their impact on customer acquisition, retention, average order value, and overall profitability, ensuring strategic alignment with business goals.

- \* Identify and document the specific operational efficiencies or favorable carrier contract terms that contribute to the unusually low shipping costs. Explore opportunities to further enhance these efficiencies and consider leveraging them as a competitive differentiator in marketing and sales.

- \* Based on the data verification and ROI analysis, evaluate alternative shipping models such as tiered pricing, minimum order values for free shipping, or adjusting product bundling strategies to optimize profitability without compromising customer satisfaction.

- \* Analyze the current product mix's influence on shipping costs and assess how potential changes in product offerings (e.g., introducing larger or heavier items) might impact future logistics expenses, allowing for proactive planning and strategy adjustments.

#### **\*\*7. Conclusion\*\***

The business demonstrates solid performance with healthy revenue and average order values. However, the unusually low average shipping cost per order represents a critical anomaly that requires immediate and thorough investigation. Understanding the true nature of this low cost – whether it's due to exceptional efficiency, strategic subsidy, or data inaccuracy – is paramount. Implementing the recommended audit and analysis will ensure accurate financial reporting, allow for informed strategic decisions regarding shipping and logistics, and ultimately contribute to sustainable profitability and operational excellence.